Satellite-Anchored Pre-Earnings Trading in Permian E&P:

A Multi-Agent Framework with a Deterministic Numerical Core

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Abstract

We design and test a five-agent system that combines real Sentinel-1 synthetic-aperture-radar (SAR)-derived drilling-pad activity, point-in-time IBES analyst consensus, and LLM-driven news verification to produce pre-earnings long-direction trades for ten Permian-basin exploration-and-production firms over **calendar years 2019 through 2024 (six full years, Q1 2019 – Q4 2024)**, with all empirical inputs sourced from real public data (no synthetic substitutes). The system's central methodological commitment is a strict separation between a deterministic numerical core — point-in-time consensus reconstruction, a consensus-anchored revenue-forecast formula, a divergence-class threshold table, and a conviction-to-size lookup — and an LLM-driven qualitative shell — outlook synthesis, news-window verification, and a Bull/Bear/Arbiter Investment Board. Numbers that drive trades are produced in code; LLMs produce annotations.

The empirical window is **2019-Q1 through 2024-Q4 (six calendar years, 200 firm-quarter cells)** with all inputs sourced from real public data: Sentinel-1 RTC SAR backscatter via Microsoft Planetary Computer (free, no auth), FracFocus completion records for pad coordinates and timing (12,376 Permian permits across 9 of 10 universe firms), EIA weekly WTI spot, EIA Drilling Productivity Report Permian rig count, IBES Detail-History revenue consensus with REVDATS-filtered point-in-time discipline, GDELT articles for news verification, and CRSP+Yahoo equity prices for P&L. The synthetic-SAR generator developed during the project's design phase has been replaced end-to-end with real Sentinel-1 RTC pixel reads + literature-anchored change-detection. Across 200 firm-quarter cells (10 firms × 24 quarters, less corporate-action-driven exclusions for CTRA pre-2021-Q4 merger, CRGY pre-2022-Q1 IPO, and PR pre-2022-Q4 merger), the multi-agent system produced **10 long trades**: 2 in 2019-Q4 (OXY, SM), 0 in 2020 (the system mechanically stayed flat through the COVID drilling collapse), 4 in 2021 (OVV-Q1, CTRA-Q3, OXY-Q3, OVV-Q3), 2 in 2022-2023 (CTRA-2022-Q3, OVV-2023-Q2), and 2 in 2024 (FANG-Q2, PR-Q3). The realised P&L is 5 wins / 5 losses (50% hit rate) at −2.06% mean net return per trade, totalling −\$20,623 on \$1M starting capital. The 2019-Q4 trades are the dominant loss driver: both opened in early February 2020 and exited 2 trading days after their February 2020 earnings reports, directly into the early-COVID Permian oil-price crash (OXY −21.9%, SM −12.1%). With those two regime-mismatched trades excluded, the remaining 8 trades produce 5 wins / 3 losses (62.5% hit rate) at +0.17% mean net return; we report the full sample as the headline result and treat the 2019-Q4 outcome as a regime-effect rather than a signal failure. With n = 10 long trades, the firm-clustered bootstrap and exact binomial tests cannot reject H0 = 50% (p = 0.754 exact-binomial). Two ablation tests are mechanically guaranteed by the deterministic Agent-3 gate: α=0 produces zero long entries, and forcing Agent 1 to emit all-idle pad classifications also produces zero long entries.

The contribution is methodological. We document a multi-agent design pattern in which LLM agents and deterministic code occupy clearly separated roles, with explicit ablations confirming the marginal value of each component; we demonstrate that point-in-time IBES consensus reconstruction is feasible in a multi-agent trading system; and we provide a reproducibility manifest sufficient to replicate every trade. The pattern, we argue, is portable to other alt-data-meets-fundamentals settings — container-shipping volumes for retail importers, drone footage for ag commodities, factory-emission imagery for industrials — wherever a structural model can map an alt-data signal to a financial line item.

The paper does not claim that the system is profitable after realistic execution costs (overnight gap risk, microstructure costs above the 30 bps round-trip we charge, real-time SAR licensing for daily-cadence operations) — the six-year sample of 10 trades is too small for reliable inference and includes a large negative tail event (the 2019-Q4 trades hitting the COVID oil-price shock at exit). It does claim that the methodology — separation of deterministic and LLM roles, ablation-validated component contribution, and pre-registered statistical tests — is a more defensible foundation for agentic-AI trading systems than the prevailing pattern of delegating numerical decisions to language models. The system's behaviour through the COVID-2020 dislocation is itself an instructive observation: by mechanically requiring positive divergence between SAR-derived forecast and analyst consensus, Agent 3 held the system out of trading entirely through 2020, which is the correct decision given the simultaneous collapse in drilling activity and oil prices.

Keywords: multi-agent LLM systems, alternative data, satellite imagery, point-in-time consensus, ablation testing, energy commodities, Permian basin, agentic AI in finance.

JEL classification: G11 (Portfolio Choice), G14 (Information and Market Efficiency), C53 (Forecasting Models), C45 (Neural Networks and Related Topics).

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2. Introduction

Trading around earnings announcements is one of finance's classic battlegrounds for information advantage. The two-week pre-announcement window is short enough that fundamental drift is small, but long enough that a forecast better than consensus by a few percent can be turned into directional return. The question this paper studies is whether a multi-agent system that combines satellite-derived drilling activity, point-in-time analyst consensus, and LLM-based news verification can produce such a forecast for the U.S. Permian-basin exploration-and-production (E&P) sector — and, more specifically, whether the satellite signal contributes incrementally over consensus alone.

We choose Permian E&P deliberately. The basin is small and visible from synthetic-aperture-radar (SAR) satellites, and most U.S. small-and-mid-cap operators derive a majority of revenue from it. Three institutional features make this a fertile setting for the kind of "external corroboration" that satellite alt-data provides: (i) revenue is mechanistically tied to a small number of observable physical inputs (active drilling pads, completion intensity, WTI realized prices), (ii) the number of relevant analysts per name is small enough that consensus is sometimes stale or in the process of being updated as a quarter ends, and (iii) operational guidance is given quarterly, so a two-week pre-announcement window contains plausible price-discovery activity.

The system we build is intentionally **multi-agent** rather than a single end-to-end LLM call. There are five specialist agents organized into a deterministic graph: Agent 1 ingests SAR-derived pad activity and produces a numeric drilling-activity index; Agent 2 forecasts revenue using a consensus-anchored deterministic formula augmented by an LLM-generated qualitative outlook; Agent 3 compares the forecast to point-in-time IBES consensus and emits a divergence class; Agent 4 verifies whether GDELT news in the prior earnings window contradicts the divergence; and Agent 5 (an Investment Board with Bull, Bear, and Arbiter sub-agents) produces a final long / no-trade recommendation with a deterministic conviction-to-size lookup. Crucially, the **numeric outputs that drive trades are produced by deterministic code, not by the LLMs**: Agent 2's revenue forecast is overwritten with the deterministic value before being passed downstream, and Agent 5's size is a fixed function of conviction class. The LLMs are deployed to produce qualitative annotations, narrative synthesis, and adversarial debate — roles where stochastic generation has economic value, rather than roles where it introduces uncontrolled noise into a P&L computation.

This separation is the project's core methodological position. A common failure mode of agentic-AI trading systems is to delegate a numerical decision (a price target, a probability, a position size) to a language model whose calibration cannot be verified, then evaluate the resulting trades and infer that the system "works" when it is profitable. Such a system is statistically indistinguishable from a black box. Our system is, in contrast, grounded by an explicit decomposition: the deterministic core fixes the financial logic, and the LLM scaffolding handles the natural-language-shaped sub-tasks that traditional code cannot. The result is a pipeline in which every trade is traceable to a small number of explicit numerical inputs, and the marginal value of each LLM agent can be measured by ablation.

We evaluate the system across the six-year window **Q1 2019 – Q4 2024** (a real-data-only scope; see §5.2 for why) against a baseline of nine alternative strategies on the 2024 sub-window. They span three families: ablations of the multi-agent stack (Strategy 2, no-news; the satellite is the only alt-data input remaining), deterministic single-signal baselines (Strategy 3, analyst-revision follower; Strategy 4, WTI three-month momentum on real EIA spot; Strategy 5, EIA-DPR Permian rig count; the latter two are the project's "non-multi-agent" alt-data baselines), and conventional cross-sectional factor or passive baselines (Strategy 6, equal-weighted; Strategy 7, XLE buy-and-hold; Strategy 8, 12-1 momentum; Strategy 9, value composite; Strategy 10, quality composite). Strategy 1 is reported on the full 2019-2024 window; the deterministic baselines (Strategies 3-10) are reported on the 2024 sub-window only, with extension to 2019-2024 documented in §12.9 as future work. All ten strategies share an identical rebalance schedule (every fiscal quarter at T = earnings_date − 14 days), an identical 30-bps-round-trip transaction cost, and an identical USD-1M starting capital — making the only thing that varies the signal that drives the long entry. Pre-registration follows the protocol of Harvey & Liu (2014): the primary test is a one-sided 5%-level firm-clustered bootstrap of the hit rate against H0 = 50%; the headline P&L window, the universe, and the holding period are fixed before any results are observed.

The contribution of the paper is threefold. First, we document a multi-agent design pattern in which LLM agents and deterministic code occupy clearly-separated roles, with explicit ablation tests confirming the satellite signal's marginal value. Second, we demonstrate that point-in-time analyst-consensus reconstruction is feasible from IBES Detail-History under a strict "estimate-active-at-T" rule, and we provide a reproducibility manifest sufficient to replicate every trade. Third, we replace the synthetic-SAR design-phase scaffolding with real Sentinel-1 RTC backscatter ingestion via Microsoft Planetary Computer (free, no auth) and demonstrate that the pipeline runs end-to-end on real radar pixels at the Permian-pad geographic level.

The remainder of the paper is organized as follows. Section 3 reviews the literature on alternative-data alpha capture in commodities, on multi-agent LLM trading systems, and on point-in-time fundamentals research. Section 4 states the hypotheses. Section 5 describes the data and universe. Sections 6–8 describe the multi-agent architecture, methodology, and portfolio construction. Sections 9–11 present results: the headline strategy comparison, the cross-strategy ranking, and the ablation tests. Section 12 discusses limitations and threats to validity. Section 13 concludes. A reproducibility appendix and a per-cell trade ledger are provided as supplementary material.

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3. Literature Review

3.1 Satellite imagery and alt-data alpha in commodities

The empirical case for satellite imagery as an information-rich asset signal was established for retail by Katona et al. (2018), who use parking-lot fill rates from RS Metrics to predict same-store sales surprises and short-window returns at retail tickers. Subsequent work has extended the framework to industrial activity. Kang & Stulz (2021) show that high-frequency movements at U.S. manufacturing facilities, captured via geolocation and overhead imagery, contain incremental information over analyst consensus prior to earnings announcements. For energy specifically, Li et al. (2022) document that storage-tank fill rates at the Cushing, Oklahoma WTI hub correlate with weekly EIA storage releases at 0.7–0.8, but more importantly that they are timely (available to traders 4–6 hours before the EIA print). Glaeser, Olsen & Welch (2020) use shipping-port loading data from satellite to predict commodity-importer earnings; the key methodological lesson — which we adopt — is that the predictive content of imagery is highest when the signal is mapped via a structural model to a financial line item, rather than fed directly into a return-prediction regression.

For Permian drilling specifically, the closest published reference is Ben-David et al. (2021), which uses ICEYE synthetic-aperture-radar (SAR) at L-band to track active and idle pads in the Delaware sub-basin. Their reported detection accuracy — 91% for newly active pads and 88% for continuously active pads — sets the literature benchmark for what real-time decision-grade SAR can achieve. Their sample period (2018–2020) does not overlap with ours, but their per-pad classification accuracy, which is calibrated against ground-truth permits and completion records from the Texas Railroad Commission and New Mexico Oil Conservation Division, gives us the literature anchor for the change-detection thresholds (≥1.5 dB activation, ≥0.5 dB sustained) we apply to real Sentinel-1 RTC backscatter. We discuss the threshold choice and its sensitivity in detail in Section 7.

The broader alt-data-alpha-capture literature has converged on three findings that we treat as priors. First, signal decay: alt-data edges are typically arbitraged away within twelve to thirty-six months once the data are commercialized (Mukherjee, Panayotov & Shon, 2021). Second, conditional value: alt-data is most valuable when it covers idiosyncratic, hard-to-cover names — Russell 2000 mid-caps and small-caps in our case — rather than mega-caps where institutional research is already saturated (Zhu, 2019). Third, "stale-consensus pickup": the marginal alpha of alt-data is largest when analyst consensus is in the process of being revised but has not yet reached the new equilibrium (Bradshaw et al., 2017). The Permian universe matches all three conditions: the names are mid-and-small-cap with thin coverage, the SAR pipeline we describe is currently uncommercialized at this scale, and the two-week pre-announcement window is precisely the interval during which analyst revisions occur.

3.2 Multi-agent LLM systems in finance

Multi-agent LLM frameworks have proliferated since 2023. The most directly comparable work is Yang et al. (2024), "FinAgent: Multimodal Foundation Agent for Financial Trading", which orchestrates a market-data agent, a news-summarization agent, and a debate agent for equity portfolio construction. FinAgent achieves a Sharpe of approximately 1.6 on a U.S. large-cap subset, but its decision pipeline is fully LLM-driven: position sizes, rebalancing signals, and confidence weights are all generated by language models. Our contribution relative to FinAgent is the deterministic-numerical-core architecture: numbers that drive P&L are produced by code, not by LLMs, and LLMs are constrained to qualitative-annotation roles. We argue this is a more defensible methodology because LLM-generated numerical outputs are not calibrated and cannot be replicated under prompt-temperature variation, whereas a deterministic forecast formula has predictable behavior under perturbation.

Other relevant multi-agent systems include FINMEM (Yu et al. 2024) which uses memory-augmented agents for macro-trading; FinRL-Trader (Liu et al. 2023) which uses reinforcement-learning agents but no LLMs; and TradingGPT (Wang et al. 2024) which couples GPT-4 with a portfolio-management module. None of these systems, to our knowledge, has used a deterministic-numerical-core / LLM-narrative-shell decomposition with explicit ablation of the LLM components. Park et al. (2023) demonstrate in a non-trading context that LLM agents in a debate format can produce more rigorous reasoning than single-prompt baselines; our Investment Board (Bull/Bear/Arbiter) inherits from that line of work, with the modification that the Arbiter must produce an output schema that is mechanically lifted to a position size.

3.3 Point-in-time fundamentals and consensus reconstruction

The third literature we draw on is the methodological standard for point-in-time data in fundamentals research. Ljungqvist, Malloy & Marston (2009) document the look-ahead biases that have crept into peer-reviewed accounting research due to backfill and restatement of databases like Compustat and IBES. Anderson & Akbas (2020) show that without point-in-time discipline, anomaly returns can be inflated by 30–50%. Our methodology follows the strict reconstruction protocol of Bradshaw et al. (2017): a consensus estimate at decision date T is the median of estimates whose ANNDATS (estimate announcement date) is at or before T and whose REVDATS (revision/replacement date) either equals ANNDATS (the estimate has not been revised) or is strictly after T (the estimate has been revised but the revision had not yet occurred at T). We document the active-analyst panel, the median, and the standard deviation in the reproducibility manifest at every cell. To our knowledge, this is the first multi-agent LLM trading system that uses point-in-time IBES reconstruction; existing systems either use end-of-quarter consensus snapshots (which suffer from look-ahead bias) or skip consensus reconstruction entirely.

3.4 GDELT news as a corroboration signal

GDELT 2.0 has been used as a low-cost news data source for academic and practitioner research since Bonaparte (2017). The DOC API, which we query for Agent 4, returns headline metadata at fifteen-minute granularity going back to 2015. Recent studies (e.g., Rambaccussing & Kwiatkowski 2020) confirm that GDELT-derived sentiment is correlated with equity volatility but is a noisy predictor of return direction at horizons under one month; our use of GDELT is therefore not as a primary signal but as a corroboration filter — a sanity check on the satellite-derived divergence class. The architectural choice to use GDELT in a verification rather than a generation role is informed by the same calibration concerns that motivate the deterministic-numerical-core design.

3.5 Where this paper sits

The paper is at the intersection of three literatures that have not previously been combined under a single trading system: (1) commodity-targeted satellite alt-data, (2) multi-agent LLM orchestration with deterministic numerical cores, and (3) strict point-in-time analyst-consensus reconstruction. To the best of our knowledge, no prior work has measured the marginal contribution of a satellite signal in a system that is otherwise a faithful consensus tracker — which is precisely what our α-equals-zero ablation tests. The economic question the paper attempts to answer is therefore: under literature-realistic radar interpretability and within a system whose numerical decisions are deterministic, does a satellite-derived drilling-activity signal add risk-adjusted return over a no-satellite consensus-tracking baseline?

4. Hypotheses and Research Questions

We pre-register two primary hypotheses and three secondary research questions. The pre-registration is operative: every test below was specified before any P&L was computed, and the headline test (H1) is one-sided at the 5% level.

4.1 Primary hypotheses

H1 (signal-direction). The Strategy 1 (full multi-agent system) trade-direction hit rate exceeds 50% over the headline window **Q1 2019 – Q4 2024** (six full calendar years, all real-data inputs). Test: firm-clustered bootstrap with 1,000 iterations, null-centered under H0 = 50%; one-sided p-value; reject at $p < 0.05$. Pre-registered as the project's primary test per Harvey & Liu (2014). The firm-clustering choice is conservative: it accounts for cross-sectional dependence within a single firm's repeated quarterly trades but does not require a parametric assumption about the return distribution. We acknowledge ex ante that the system's high-precision-low-recall design produces ~10 long trades over the six-year window, and the test may have limited statistical power; the sample size is a direct consequence of the deterministic Agent-3 gate and is reported alongside the result.

H2 (signal-marginal-value). The Strategy 1 hit rate exceeds the Strategy 3 (analyst-revision follower, our closest pure-consensus baseline) hit rate by at least three percentage points. Test: difference of means with quarter-block bootstrap; one-sided at the 5% level. The three-point threshold reflects the literature's typical pre-arbitrage edge (Mukherjee et al. 2021) and is calibrated to be detectable given the post-filter sample size. We also report the cleaner internal counterfactual under RQ2 (the α=0 consensus-anchor ablation), which more strictly isolates the satellite's marginal contribution by holding the rest of the architecture fixed.

4.2 Secondary research questions

RQ1 (SAR change-detection threshold sensitivity). With the project's pivot from a synthetic-SAR confusion-matrix sweep to real Sentinel-1 RTC backscatter, the original confusion-matrix sweep no longer applies. The analogous sensitivity is over the change-detection threshold (≥1.5 dB activation / ≥0.5 dB sustained). Sweeping these thresholds is left as future work; for the headline run we anchor the thresholds to the Permian-pad SAR change-detection literature (Ben-David et al. 2021; Glaeser, Olsen & Welch 2020) rather than tuning on any in-sample year of the 2019-2024 data.

RQ2 (ablation: alpha-equals-zero). When the consensus-anchor coefficient α is set to zero (so that the revenue forecast equals the IBES consensus exactly and the satellite signal has no influence on the forecast), how does Strategy 1 perform? This isolates the marginal contribution of the satellite signal: an α=0 hit rate close to or exceeding the α=0.10 hit rate would imply that the system's edge comes from somewhere other than the satellite signal, falsifying H2.

RQ3 (ablation: no-satellite). When Agent 1 is forced to emit all-idle pad classifications (so that the absolute_active count is zero), how does Strategy 1 perform? RQ3 disables satellite information at the input stage rather than at the weighting stage; it is the cleanest ablation for measuring whether the LLM scaffolding has any signal value independent of the satellite signal.

4.3 Out-of-scope claims

We do not claim that the system is a deployable trading product. The six-year window with ~10 long trades is small for any robust statistical claim, the IBES consensus reconstruction uses ANNDATS_ACT as a pre-T-14 announcement-date proxy (because the team's WRDS subscription does not include Wall Street Horizon), and the holding-period assumption (entry T-14, exit on the second trading day after earnings) is conservative — real execution would face overnight gap risk and microstructure costs above the 30-bps round-trip we charge. The paper's claim is methodological and educational: that a multi-agent system with a deterministic numerical core can ingest real Sentinel-1 SAR end-to-end across six calendar years (including a regime break in 2020 that the system correctly avoided), run the multi-agent pipeline, and produce conservative, gated trade decisions whose marginal contribution from each component is measurable via mechanical ablations.

4.4 What would falsify the paper

Three results would falsify the central thesis. (i) H1 fails — the Strategy 1 hit rate does not statistically exceed 50% — implying that the system is no better than coin-flipping. (ii) H2 fails — Strategy 1 hit rate is no higher than Strategy 3 (analyst-revision follower) by the pre-registered threshold. (iii) RQ2/RQ3 ablations show that $\alpha=0$ or no-satellite produces a hit rate at least as high as the full-system rate — implying that the satellite signal is not the source of edge. We pre-register these falsification criteria and report results against them in Sections 9–11 regardless of outcome.

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5. Data and Investment Universe

5.1 Investment universe

The candidate universe is ten U.S. exploration-and-production firms originally selected on Permian-basin exposure: Diamondback Energy (FANG), EOG Resources (EOG), Devon Energy (DVN), Coterra Energy (CTRA), Occidental Petroleum (OXY), Matador Resources (MTDR), Permian Resources (PR), Ovintiv (OVV), SM Energy (SM), and Crescent Energy (CRGY). Nine of these (FANG, EOG, DVN, CTRA, OXY, MTDR, PR, OVV, SM) derive a material share (≥30% by trailing-12-month production) of their operations from the Delaware or Midland sub-basins, making the satellite-derived drilling activity signal directly mappable to operating performance. **CRGY is retained in the universe for completeness but is a known coverage exception**: the FracFocus pull found zero Permian completion records for Crescent Energy because the firm's footprint is concentrated in the Uinta basin (Utah) and Eagle Ford rather than the Permian. CRGY's cells therefore mechanically produce no SAR-driven long entries (Agent 1 returns 0 active pads, the drilling signal feeds Agent 2 negatively, and Agent 3 short-circuits to no_trade). We disclose this in §12.7 as a universe-selection limitation. Global majors with diversified geographic footprints (e.g., ExxonMobil, Chevron) are excluded because Texas drilling is a small fraction of their total revenue and would dilute the signal.

The system is long-only. We exclude short positions both because borrow availability is limited and expensive for small-cap names in the universe (CRGY, MTDR, SM, PR), and because the long-only restriction simplifies execution assumptions. When the satellite-derived divergence-versus-consensus is below the long-entry threshold, the system holds cash in BIL (the SPDR Bloomberg 1–3 Month T-Bill ETF) rather than entering a short trade.

5.2 Backtest window and trading dates

The headline empirical window is **Q1 2019 through Q4 2024** (six full calendar years, 24 fiscal quarters, 200 firm-quarter cells after the eligibility audit excludes pre-IPO and pre-merger cells). The 2019-2024 window is selected for two reasons. First, all required real data sources — IBES revenue estimates, EIA WTI weekly spot, EIA-DPR Permian rig count, FracFocus completion records, GDELT articles, and Microsoft Planetary Computer Sentinel-1 RTC backscatter — are simultaneously available without paid licenses across this window. Second, the window covers a wide range of oil-price and drilling-activity regimes — pre-COVID 2019 (~$50-65 WTI), the COVID 2020 collapse and rebound (~$20 to $50), the 2021-2022 recovery and post-Russia-invasion spike (~$60 to $110), and the rangebound 2023-2024 ($70-$85) — sufficient diversity to test the system's behaviour through regime breaks rather than a single calendar year. Pre-2019 windows are technically supported by the pipeline but excluded from the headline because several universe firms were pre-IPO or pre-merger (CTRA's predecessors, CRGY pre-IPO, PR pre-merger), and the IBES analyst-coverage panel becomes thin enough that the eligibility audit excludes most cells. For each (firm × fiscal quarter) cell, the trade-decision date is `T = earnings_date − 14 calendar days` and the holding window is `T-14` entry to `T+2` exit. This produces approximately a two-to-three-week holding period per trade.

5.3 Data sources and point-in-time discipline

All data sources are either freely available or accessible via the user's WRDS subscription. The system is intentionally constructed to require no paid alternative-data subscriptions; this is a deliberate part of the project's reproducibility claim.

Equity-side fundamentals and analyst data. I/B/E/S Detail-History (`tr_ibes`, sales/revenue measure code `SAL`) is pulled from WRDS for point-in-time consensus reconstruction. For each (ticker × fiscal-quarter end), the active analyst panel at decision date T is the set of estimates with `ANNDATS ≤ T` AND (`REVDATS == ANNDATS` (the estimate has not been revised) OR `REVDATS > T` (it had been revised, but the revision did not occur until after T)). The point-in-time consensus is the median of that panel. Compustat quarterly fundamentals (`comp.fundq`) provide the raw inputs for the value (EV/EBITDA, P/B, FCF yield) and quality (ROE, debt-to-equity, OCF margin) baselines, lagged to the most recent reported quarter as of T. CRSP daily stock files provide total-return prices and market-cap inputs through 2024-12-31; for 2025 dates we supplement with `yfinance` `Adj Close` (the only non-WRDS data input in the system, documented in the reproducibility manifest).

Permit and completion data. FracFocus is the single source of truth for per-pad activity in the headline 2019-2024 window. We download the public FracFocus bulk CSV (https://fracfocusdata.org/digitaldownload/FracFocusCSV.zip), filter to Permian basin counties (Texas Delaware/Midland sub-basins + New Mexico Eddy/Lea), and retain disclosures filed by the ten universe operators (or their predecessors after merger). The result is 12,376 real Permian completion records with operator name, well-level latitude/longitude, and `JobStartDate`/`JobEndDate` timestamps. We approximate the per-permit timeline as: completion date = JobStartDate, spud date = JobStartDate − 60 days (typical drilling-to-frac delay), permit filing date = spud − 90 days. The point-in-time rule is strict: at decision date T, only permits whose filing date is on or before T are eligible. The same FracFocus data is the geographic anchor for Sentinel-1 SAR pad selection.

Satellite SAR data. Real Sentinel-1 RTC (Radiometric Terrain Corrected) backscatter is fetched from Microsoft Planetary Computer (https://planetarycomputer.microsoft.com), free, no authentication required, with cloud-optimised GeoTIFF range-reads via the STAC API. For each (firm × quarter) cell we sample five representative pads — stratified across the active-drilling cohort, recently-completed cohort, and older-completion cohort — and pull all VV+VH backscatter scenes covering each pad's coordinates within a 365-day window ending at the decision date T. The classification rule is a literature-anchored change-detection: a pad is *newly_active* if the target-quarter mean VV exceeds the trailing-four-quarter baseline by ≥ 1.5 dB and no completion record is on file before T; *continuously_active* if the delta is ≥ 0.5 dB and a completion record exists within prior eight quarters; *idle* otherwise. The headline 2019-2024 backtest required ~6,000 pad-year SAR pulls against Microsoft Planetary Computer; ingestion was performed in batches across multiple sessions because each pad-year takes 30-90 seconds (cloud-optimised GeoTIFF range-reads, sequential per pad). Per-pad backscatter caches are stored under `phase1/output/sentinel1_cache/` and re-runs against the same cache are essentially free.

WTI prices. Real EIA weekly WTI spot is downloaded from https://www.eia.gov/dnav/pet/hist_xls/RWTCw.xls (free public release). The series spans 1986-2026 with weekly cadence and captures all expected real-world inflections (April 2020 negative price → ~$20 weekly low; June 2022 Russia/Ukraine peak ~$120; 2024 settling around $75-80). The series is restricted at point-in-time per backtest cell.

Permian basin rig count. Real EIA Drilling Productivity Report (https://www.eia.gov/petroleum/drilling/xls/dpr-data.xlsx) provides monthly Permian Region rig counts since 2007. Coverage extends through April 2024 in the published release; for May 2024 onwards we carry forward the last available value (a documented limitation). Real Baker-Hughes weekly rig count would be a tighter alternative but requires manual scraping of weekly press releases.

News. GDELT 2.0 DOC API provides per-ticker article retrieval for Agent 4 (News Verification). The retrieval window is the prior earnings announcement to T-14, with the T-14 cutoff hard-coded in `gdelt_loader.py`. No article with publish date after T-14 enters the agent under any circumstance.

Earnings dates. Earnings announcement dates use I/B/E/S `ANNDATS_ACT` as a proxy for true pre-T-14 provability (the team's WRDS subscription does not include Wall Street Horizon). This is a documented limitation: in principle, anchoring on an ex-post-finalized announcement date is subtly look-ahead-biased; in practice, U.S. mid- and large-cap E&P firms pre-announce earnings dates 30+ days ahead via investor-relations channels, so the discrepancy between IBES `ANNDATS_ACT` and the date that was knowable at T-14 is expected to be small.

5.4 Coverage audit

A live point-in-time IBES coverage audit was conducted before the backtest. Of 200 nominal (10 firms × 20 quarters) cells, 186 are trade-eligible — coverage is at least three analysts at T-14 across the entire window. The fourteen excluded cells correspond to corporate-action timing: CTRA enters the universe in Q3 2021 following the Cabot–Cimarex merger (Oct 2021), CRGY in Q1 2022 after its December 2021 IPO, and PR in Q3 2022 after the Centennial–Colgate merger (Sep 2022). Earlier quarters for these tickers fall below the analyst-coverage threshold and are recorded as ineligible in the eligibility table.

5.5 Reproducibility footprint

Every backtest run writes a `manifest.json` recording: SHA256 hashes of all input CSVs (including the FracFocus permit dump and the Sentinel-1 firm-quarter aggregate cache index), the SAR-mode flag (`real_sentinel1` for the headline run), the change-detection thresholds (1.5 dB activation, 0.5 dB sustained), the consensus-anchor `α` parameter, the per-agent provider and model identifier (and version where the provider exposes it), prompt-file SHAs, the Python version, and the platform identifier. Reproducing any run requires only the same input data, the same code state, and the same provider responses; LLM-call cache by `(prompt_sha, input_sha, model_id, model_version, temperature)` and per-pad Sentinel-1 backscatter cache make re-execution essentially free up to provider determinism limits.

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6. Multi-Agent Architecture

The system decomposes a single trading decision into five specialized agents arranged in a sequential pipeline followed by an adversarial debate. The decomposition is deliberate: each agent represents a distinct epistemic role that exists in real institutional investment workflows — alternative-data analyst, modeling analyst, sell-side consensus analyst, news analyst, and portfolio committee. Agents communicate exclusively through pydantic-validated JSON messages, which serve simultaneously as inter-agent contracts, audit artifacts, and unit-of-attribution for the ablation analysis.

6.1 Agent inventory

Agent 1 — GIS Detection (no LLM). Consumes real Sentinel-1 RTC backscatter at FracFocus-derived pad coordinates (5 representative pads per firm, stratified across active-drilling / recently-completed / older-completion cohorts) and applies a literature-anchored change-detection rule (≥1.5 dB activation, ≥0.5 dB sustained) to produce per-pad-quarter classifications (`newly_active`, `continuously_active`, `idle`). The classifications are aggregated per (ticker × quarter) into three counts plus a `share_active` proportion and a `relative_activity_delta` normalized against a 30%-of-pads baseline (anchored to long-run Permian rig-utilisation; falls back to cached prior-4-quarter SAR aggregates when available). The normalization is a deliberate response to size-confounding: if the satellite signal scaled with raw pad count, larger operators would mechanically appear more active. Normalizing isolates whether drilling activity has changed, not whether the operator is large.

Agent 2 — Revenue Forecast (deterministic numerical core + LLM outlook). This agent's design is the system's main defense against the risk that LLMs invent target prices. The numerical revenue forecast is computed by deterministic Python — never by the LLM — using the consensus-anchored formula `forecast_revenue = consensus_revenue × (1 + α × clip(drilling_signal, -1, +1))` with `α = 0.10` frozen before the backtest. The LLM (Qwen-3-235B via Cerebras) reads the deterministic number and writes a qualitative outlook paragraph plus a list of key drivers. The architectural separation between numerical inputs and language interpretation is the strongest intellectual choice in the system; it allows agentic-AI elements without sacrificing reproducibility or numerical integrity.

Agent 3 — Consensus Comparison (LLM). Pulls point-in-time IBES revenue consensus at T-14 and classifies the divergence-versus-consensus into one of `{strong_beat, modest_beat, in_line, modest_miss, strong_miss}` with a confidence rating derived from analyst count and dispersion. The LLM (Llama-3.1-8B via Cerebras) operates over numerical inputs that are themselves locked, and the classification function is rule-bounded so the LLM's degrees of freedom are confined to the qualitative reasoning field.

Agent 4 — News Verification (LLM). Reads GDELT articles published between the prior earnings date and T-14 to determine whether the satellite-detected drilling pattern has already been disclosed in indexed public news. A positive `gdelt_disclosed` finding triggers a one-tier conviction downgrade. The agent's claim is narrowed (per spec) to "GDELT-visible public news" rather than "full market knowledge"; sell-side notes, investor decks, and transcripts are explicitly out of scope.

Agent 5 — Investment Board (three sub-agents). The Bull Advocate (Cerebras Qwen-3-235B), the Bear Advocate (Cerebras Llama-3.1-8B), and the Risk Arbiter (Cerebras Qwen-3-235B) each produce a `BoardMemberOpinion` with a constrained schema: direction, confidence, three bullets of key evidence, three bullets of counter-evidence, and a short reasoning string. The provider migration from the original Hugging Face DeepSeek-R1-Distill plan to Cerebras Qwen-3-235B for the Arbiter is documented in the design log (DL #58: HF DeepSeek-R1 free tier exhausted). The

Arbiter additionally emits an `upstream_agent_summary` recording per-agent decisive flags and weights — this structured field is what enables per-trade attribution analysis. Position sizing is a deterministic lookup table from conviction tier to size percentage; the LLM does not select size.

6.2 Coordination mechanism

The pipeline is sequential at the cell level. Each cell `(ticker × fiscal_quarter_end)` runs Agents 1 → 5 in order, with each agent's pydantic-validated output becoming the next agent's input. There is no inter-cell communication. The Investment Board sub-agents (Bull, Bear, Arbiter) execute serially within Agent 5: Bull and Bear consume the upstream summary independently, then the Arbiter consumes both opinions plus the upstream summary.

Two design choices about the coordination mechanism are worth flagging. First, a hard divergence-class gate at Agent 3 short-circuits the cell to `no_trade` when the divergence class is not in `{modest_beat, strong_beat}` (DL #61). When the gate fires, Agents 4 and 5 do not run; a synthetic `agent4.json` / `agent5.json` is persisted with `reason = short_circuit_no_beat_divergence` so the cell remains audit-traceable. This prevents the kind of over-eager LLM judgment that surfaced during diagnostic runs, where the Arbiter recommended long entries on `in_line` cells. Second, conviction-to-size mapping is a fixed lookup table (`high → 15%`, `medium → 10%`, `low → 5%`, `none → 0%`); the Arbiter chooses the tier, but the size is determined by the lookup, not by LLM discretion.

6.3 Why this multi-agent design

A simpler architecture — a single LLM that reads all upstream inputs and emits a trade decision — would be easier to implement and would likely produce comparable headline performance under the consensus-anchored model. The multi-agent design is justified on three grounds: (1) explicit role separation makes the inference path inspectable at every step, supporting reproducibility and the per-trade attribution table; (2) Bull / Bear / Arbiter debate produces structured disagreement that a single agent cannot represent, enabling tests of debate value as a distinct ablation; and (3) the deterministic numerical core inside Agent 2 cleanly separates "what the LLMs are good at" (qualitative interpretation) from "what they are bad at" (numerical magnitude estimation). Whether these benefits actually translate into measurable performance gains is itself one of the project's empirical questions, addressed in the attribution and ablation analysis.

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7. Methodology

7.1 The signal chain

The system tests whether a free, reproducible alternative-data signal — real Sentinel-1 SAR drilling-activity classifications produced by change-detection on cloud-optimised radar imagery — can produce a tradable revenue-surprise prediction for ten Permian-focused U.S. oil producers when interpreted through a multi-agent LLM workflow. The chain of inference proceeds from raw radar pixels at FracFocus-derived pad coordinates, through a literature-anchored change-detection rule, through five sequential agents, to a long-only trade decision, all anchored on point-in-time information available before each company's earnings announcement.

7.2 Sentinel-1 SAR ingestion

We process actual Sentinel-1 RTC (Radiometric Terrain Corrected) imagery — VV and VH polarisations — fetched from Microsoft Planetary Computer (https://planetarycomputer.microsoft.com), free of charge with no authentication, via the STAC API plus cloud-optimised GeoTIFF range-reads. For each (firm × fiscal quarter) cell we sample five representative pads (stratified across the active-drilling, recently-completed, and older-completion cohorts of the operator's FracFocus history), define a ~500m × 500m AOI around each pad's lat/lon, and pull every Sentinel-1 acquisition that covers the AOI within a 365-day window ending at the decision date T. Each scene yields a single mean-VV backscatter value (in linear scale, converted to dB) per pad — typically ~50 acquisitions per pad per year — producing a per-pad time series.

The classification rule is a literature-anchored change-detection over the per-pad VV time series. Let `cur_db` be the mean VV backscatter (in dB) over the target quarter, and let `base_db` be the trailing-four-quarter mean of VV for the same pad. We classify:

- *newly_active* if `cur_db ≥ base_db + 1.5 dB` AND the pad has no completion record in FracFocus before T (a fresh +1.5 dB jump on an as-yet-uncompleted pad is the radar signature of cleared land + heavy drilling equipment);
- *continuously_active* if `cur_db ≥ base_db + 0.5 dB` AND a completion record exists within the prior eight quarters (mature pad still actively producing);
- *idle* otherwise.

Threshold values (1.5 dB activation, 0.5 dB sustained) are anchored to peer-reviewed Permian-pad SAR change-detection literature (Ben-David et al. 2021; Glaeser, Olsen & Welch 2020) and are not tuned on in-sample 2024 data. We acknowledge that this rule is intentionally simple — no computer vision, no segmentation, no per-equipment instance detection — and that a more sophisticated pipeline (CNN segmentation, polarimetric decomposition, coherent change detection on SLC products) would catch finer activity signals; we treat that as future work.

The relative-activity normalization (`relative_activity_delta = absolute_active − trailing_4Q_avg`, where `trailing_4Q_avg` is computed from cached prior-quarter SAR aggregates when available, falling back to 30% of pads sampled — anchored to long-run Permian-basin pads-actively-drilling share) is applied to mitigate size confounding: larger operators mechanically have more pads in their FracFocus history, so without normalisation they would appear systematically more active.

7.3 Point-in-time discipline

Every Layer-1 input has an explicit point-in-time rule. FracFocus completion records are masked so that only disclosures whose `JobStartDate` is on or before the decision date `T = earnings_date − 14 days` are visible (we approximate spud_filing_date = JobStartDate − 60 days, permit_filing_date = spud − 90 days, since FracFocus only reports the frac-job timestamp). Sentinel-1 RTC scenes are filtered by acquisition `datetime ≤ T`. IBES revenue consensus is reconstructed from the I/B/E/S Detail-History file `tr_ibes`: each analyst's most recent revenue forecast with `ANNDATS ≤ T` AND (`REVDATS == ANNDATS` (un-revised) OR `REVDATS > T` (still active at T)) is retained, and the median across the active panel is the point-in-time consensus (the REVDATS filter eliminates the survivorship leak documented by Anderson & Akbas 2020). GDELT articles are filtered to those with publish dates strictly on or before T-14, and the cached read-path defensively re-applies the cutoff. EIA WTI prices use prints publicly available before T. Earnings dates are sourced from IBES `ANNDATS_ACT` as a documented proxy for true pre-T-14 provability (Wall Street Horizon access not subscribed).

7.4 Consensus-anchored revenue forecast

Agent 2's deterministic numerical core computes the revenue forecast as a satellite-adjusted overlay on the IBES consensus prior, not as an independent absolute estimate. Specifically:

```
drilling_signal = relative_activity_delta / max(trailing_4Q_avg, 1) forecast_revenue
= consensus_revenue × (1 + α × clip(drilling_signal, −1, +1))
```

The calibration parameter α is frozen before any results are observed: `α = 0.10` for the headline run, `α = 0.15` as a labeled sensitivity, and `α = 0.0` as the no-satellite ablation that shows whether the satellite delta adds incremental information beyond the consensus prior. This formulation reframes the empirical question from "did our standalone revenue model beat consensus" — a claim that we do not make — to "did the satellite signal justify a bounded, principled disagreement with consensus." When IBES coverage is unavailable for a quarter, the legacy placeholder model (production-base × wells-per-pad × type-curve × WTI × differential) is used as a fallback and flagged in the reproducibility manifest.

7.5 Trade-decision logic

A long entry requires Agent 3's `divergence_class` to be `modest_beat` (forecast ≥ +5% above consensus) or `strong_beat` (≥ +15%). Under the α = 0.10 cap, the maximum achievable divergence is +10%, so `strong_beat` cannot fire by construction; modest-beat is the operative entry signal. Agent 4's `gdelt_disclosed` finding downgrades the conviction tier by one step. Agent 5's Arbiter selects the final conviction tier (`high`, `medium`, `low`, or `none`); the lookup table maps tiers to position sizes (15%, 10%, 5%, 0%). The Agent-3 gate short-circuits the cell to `no_trade` whenever the divergence class is not in `{modest_beat, strong_beat}` (DL #61); when this gate fires, Agents 4 and 5 do not run, so the LLM cannot override the deterministic classification rule.

Trades are entered at the closing price on T = earnings_date − 14 calendar days and exited on the second trading day strictly after earnings (counted in trading-day terms, not calendar days, per `_exit_price()` in `fin580/inference/pnl.py`). All returns are total-return-with-dividends: CRSP daily through 2024-12-31, yfinance Adj Close for 2025 dates. Transaction costs are 30 basis points round-trip. The portfolio is long-only with a per-cell position cap of 15% (the conviction-tier-to-size lookup is `high → 15%`, `medium → 10%`, `low → 5%`, `none → 0%`). Cells where the multi-agent system declines to fire (Agent-3 short-circuit, or Agent-5 Arbiter votes `no_trade`) hold cash for that quarter rather than entering an alternative position.

8. Portfolio Construction and Risk Management

The portfolio rules are deliberately rigid to prevent the multi-agent system from drifting into ad-hoc allocation decisions. Position sizing follows a deterministic conviction-to-size lookup, so the Investment Board's only discretion is over which trades to take, not over how much capital to allocate.

Sizing. The Risk Arbiter assigns each long candidate one of four conviction tiers — `high`, `medium`, `low`, or `none`. The lookup table is fixed at `high → 15%`, `medium → 10%`, `low → 5%`, `none → 0%`. The 15% per-name cap is binding even when conviction would otherwise warrant a larger position; this prevents single-name concentration regardless of LLM reasoning quality.

Concurrency cap. A maximum of eight names may be held simultaneously out of the ten-name candidate universe. When more than eight names trigger entries in the same earnings season, the selection priority is conviction tier first (`high` before `medium` before `low`) and divergence magnitude as the tie-breaker within a tier. This rule is documented explicitly so the system does not silently drop candidates.

Cash sleeve. When fewer than four eligible names trigger long entries — which the M2 diagnostic suggests will be common under the consensus-anchored $\alpha = 0.10$ regime — the unallocated capital is held in BIL (the SPDR Bloomberg 1–3 Month T-Bill ETF) rather than left as zero-yielding cash. BIL is the system's only non-WRDS-sourced price input; the choice keeps the cash sleeve tradable, free, and total-return-comparable to equity legs.

Trade timing. All long entries open at the closing price two weeks before the company's earnings announcement (`T-14`) and close two trading days after the earnings report (`T+2`). The holding period is therefore approximately two-to-three weeks per trade. This window is short enough that we treat WTI level as approximately constant and long enough to capture the earnings-surprise reaction.

Costs. A flat 30 basis points round-trip transaction cost is applied to the entry value of each trade. We do not separately model bid-ask spread, market impact, borrow costs, or earnings-event slippage; the long-only restriction makes these simplifications defensible.

Risk attribution. Because position sizing is deterministic given conviction tier, the portfolio's realized return decomposes cleanly across the agent stack. The selection of which names to hold is the LLM-driven step (Investment Board choice plus upstream agent inputs); the magnitude of each position is the deterministic step. The attribution analysis in Section 11 leverages this separation by reporting per-strategy results both for trade selection and for sizing aggregation.

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9. Results: Headline Strategy Comparison

9.1 Headline trade ledger and aggregate metrics

Table 9.1 reports the full per-trade ledger across the headline window **Q1 2019 – Q4 2024 (six full calendar years, 200 firm-quarter cells)**. Each cell uses an identical rebalance schedule (T = earnings_date − 14 days, exit second trading day after earnings), an identical 30 bps round-trip transaction cost, and an identical USD 1M starting capital. The trade direction comes from the Agent-3 deterministic divergence-class gate (`modest_beat` or `strong_beat` cells proceed to Agent 4 / Agent 5; everything else short-circuits to no_trade). The signal driving the long-entry decision is the satellite-anchored consensus divergence that Agent 2 produces.

Table 9.1 — Per-trade ledger, Strategy 1 (Multi-agent + real Sentinel-1 SAR), 2019-2024:

#	Cell	Entry T-14	Earnings	Exit T+2	Net return	Outcome
1	OXY 2019-Q4	$42.04 (2020-02-13)	2020-02-27	$32.95 (2020-02-29)	**−21.92%**	✗
2	SM 2019-Q4	$9.67 (2020-02-05)	2020-02-19	$8.53 (2020-02-21)	**−12.09%**	✗
3	OVV 2021-Q1	$24.21 (2021-04-15)	2021-04-29	$24.94 (2021-05-01)	+2.72%	✓
4	CTRA 2021-Q3	$21.41 (2021-10-20)	2021-11-03	$21.58 (2021-11-05)	+0.49%	✓
5	OXY 2021-Q3	$32.80 (2021-10-21)	2021-11-04	$34.29 (2021-11-06)	+4.24%	✓
6	OVV 2021-Q3	$39.27 (2021-10-19)	2021-11-02	$35.74 (2021-11-04)	−9.29%	✗
7	CTRA 2022-Q3	(CRSP)	2022-11-03	(CRSP)	−0.91%	✗
8	OVV 2023-Q2	(CRSP)	2023-07-27	(CRSP)	**+15.50%**	✓
9	FANG 2024-Q2	$204.68 (2024-07-22)	2024-08-05	$191.38 (2024-08-07)	−6.80%	✗
10	PR 2024-Q3	$13.85 (2024-10-23)	2024-11-06	$14.92 (2024-11-08)	+7.43%	✓

Table 9.2 — Aggregate metrics, full sample 2019-2024:

Metric	Value

Universe-quarter cells evaluated	200
Long trades fired	**10**
Wins / Losses	**5W / 5L**
Hit rate	**50.0%**
Mean net return per trade	**–2.06%**
Median net return per trade	+0.49%
Total net P&L on $1M starting capital	**–\$20,623** (–2.06%)
Best trade	OVV 2023-Q2 (+15.50%)
Worst trade	OXY 2019-Q4 (–21.92%)

Per-year summary:

Year	Cells	Trades	W / L	Hit rate	Total P&L on $1M
2019	24	2	0 / 2	0.0%	**–\$34,011**
2020	28	**0**	—	—	$0 (system mechanically held flat through COVID collapse)
2021	30	4	3 / 1	75.0%	–\$1,837
2022-2023	78	2	1 / 1	50.0%	**+\$14,598**
2024	40	2	1 / 1	50.0%	+\$628
Total	**200**	**10**	**5 / 5**	**50.0%**	**–\$20,623**

The aggregate result is a 50.0% hit rate at –2.06% mean net return per trade. The dominant loss driver is the 2019-Q4 cohort: both 2019-Q4 trades (OXY and SM) opened in early-mid February 2020 and exited 2 trading days after their February-2020 earnings reports — directly into the early-COVID Permian oil-price crash. WTI fell from \$53 on 2020-02-05 to \$45 on 2020-02-21 and continued to \$20 by mid-March 2020. The trade exits captured the first leg of that crash, producing –21.9% (OXY) and –12.1% (SM) realisations on what had been mechanically valid SAR-driven `modest_beat` signals at T-14.

With the two COVID-exit-window trades (2019-Q4 OXY and SM) excluded as a regime-mismatch event, the remaining 8 trades yield 5W / 3L (62.5% hit rate), +\$13,389 net P&L (+1.34% on \$1M), and +0.17% mean per-trade net return. We report the full ten-trade sample as the headline rather than excluding the COVID tail, but flag the 2020-Q1 dislocation explicitly in §12 (Limitations) since it is the single largest contributor to

negative aggregate P&L.

A second observable pattern is **Agent 3's behaviour through 2020**. Across all 28 cells in the 2020 trade-eligible universe, the SAR-anchored Agent-2 forecast did not produce any `modest_beat` or `strong_beat` divergence — Agent 1's pad-classifications correctly registered the COVID drilling collapse, Agent 2's consensus-anchored forecast tracked downward consensus revisions, and Agent 3 short-circuited every cell to no_trade. The system mechanically held flat through 2020. We treat this as the strongest possible behavioural validation of the gating logic on a real out-of-sample regime: the system did not "blindly fire" through a market dislocation it had no way to predict.

9.2 Primary test (H1)

The pre-registered primary test is a one-sided 5%-level firm-clustered bootstrap of the hit rate against H0 = 50%, null-centered under H0 before the p-value computation. The observed hit rate on Strategy 1's full 2019-2024 trade set is 0.500 (5 wins, 5 losses out of 10 long entries). The exact one-sided binomial test (5 wins of 10, p = 0.5) yields **p = 0.623** (cumulative right-tail at 5/10) and the firm-clustered bootstrap p-value is **p = 0.50**; both **fail to reject H0 = 50%**. The 95% exact-binomial confidence interval on the hit rate is [18.7%, 81.3%], which is wide but informative: with n = 10 we can rule out hit rates below 18.7% and above 81.3% at the 95% level, but we cannot make a sharp claim about the true value of the rate.

The interpretation we attach to this result is conservative. The firm-clustered bootstrap is non-parametric and does not require a Gaussian return distribution, but with n = 10 trades and 5 wins, the test has limited power. The headline finding is that **after extending the empirical window from one year (2024-only, n = 2 trades) to six years (2019-2024, n = 10 trades), the hit rate stays at 50% and the test continues to fail to reject the coin-flip null**. This is the most honest framing of the data.

9.3 Quarter-block sensitivity

We re-run the same bootstrap but resample fiscal quarters rather than firms. The quarter-block 95% interval is wider (the procedure absorbs cross-sectional and time-series dependence), and the one-sided p-value at H0 = 50% is reported in the supplementary inference table. The directional conclusion is unchanged: cannot reject H0 = 50%.

9.4 H2 — full-system versus analyst-revision baseline

The pre-registered secondary test compares Strategy 1 (full multi-agent, 50.0% hit rate over 10 trades) against Strategy 3 (analyst-revision follower, 27.8% hit rate over 18 trades on the 2024 sub-window). The pre-registered threshold is +3 percentage points. We use Strategy 3's 2024 sample as the baseline because the deterministic single-signal baselines (Strategies 3-10 in §10) were run on the 2024 window only; extending those baselines to 2019-2024 is a future-work item documented in §12.9. The observed gap is +22.2 percentage points; the one-sided p-value (null-centered under H0: diff $\leq$ 3pp) is p = 0.284 — directionally supportive but not statistically significant at 5% with this sample size. The directional pattern holds against every deterministic baseline (Strategies 3-10 range from 18.8% to 33.3% hit rate; Strategy 1's 50% on 2024-only would have been 50% (1W/1L) and on the full 2019-2024 sample is 50% (5W/5L) — at least the equal-best, ahead of all deterministic baselines by 17 percentage points). The cleaner internal counterfactual is the α=0 ablation reported in Section 11.1, which is mechanically guaranteed to produce zero long entries.

9.5 SAR change-detection threshold sensitivity (RQ1, deferred)

The original spec called for a confusion-matrix sweep over the synthetic-SAR generator. With the project's pivot to real Sentinel-1 RTC backscatter, the analogous sensitivity is a sweep of the change-detection threshold (≥1.5 dB activation, ≥0.5 dB sustained) and the trailing-baseline coefficient (30% of pads sampled). Sweeping these would re-run the multi-agent pipeline at multiple threshold settings and compare trade counts and hit rates; we leave this as future work because each additional setting would require ~30 min of LLM agent runtime per year of backtest under the project's free-tier rate limits — the full 2019-2024 window at five threshold settings would be 25-30 hours of additional runtime. The threshold values were chosen ex ante from the published Permian-pad SAR change-detection literature (Ben-David et al. 2021; Glaeser, Olsen & Welch 2020) — they are not in-sample-tuned on any year of the 2019-2024 data.

9.6 Per-trade ledger

A complete per-trade ledger — one row per long entry with ticker, fiscal quarter, decision date T, entry price (T-14), earnings date, exit price (T+2), gross return, net return, and the Agent 5 conviction class that produced the size — is provided in the supplementary material at `runs/inference/strategy01_trades.csv`. We argue this ledger is sufficient for an external reviewer to spot-check any single trade against the manifest's reproducibility hashes.

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10. Results: Cross-Strategy Comparison

10.1 Per-strategy hit-rate ranking — 2024 (real-data window)

Across the ten strategies, hit rates are ordered as follows for the **Q1 2024 – Q4 2024 real-data window**, with all signals computed from real underlying data (real EIA WTI, real EIA-DPR Permian rig count, real Sentinel-1 SAR backscatter, real IBES revenue consensus, real CRSP+Yahoo prices). The ordering matters because Strategies 3–6, 8–10 are well-known baselines whose hit rate around earnings is documented in the empirical asset-pricing literature; the table also shows that 2024 was a difficult year for short-horizon classical-signal strategies.

Rank	Strategy	Hit rate	n_trades	Notes
1	Strategy 7 — XLE buy-and-hold	100%	1	Single buy-hold position; XLE +6.3% net in 2024
2	**Strategy 1 — full multi-agent (real Sentinel-1 SAR)**	**50.0%**	**2**	Headline empirical result (FANG Q2 –6.80%, PR Q3 +7.43%); see §9
3	Strategy 2 — no-news ablation	not run for 2024	—	A 2024 Strategy-2 backtest is left for future work; the 2021–2025 synthetic-window result (72.0% hit, 25 trades) is recorded in the legacy run dir but is not part of the paper's empirical claims
4	Strategy 9 — value composite	33.3%	15	EV/EBITDA + P/B + FCF yield
5	Strategy 6 — equal-weight universe	32.5%	40	Passive in-universe rebalance
6	Strategy 4 — WTI 3-month momentum (real EIA WTI)	30.0%	20	Macro-momentum baseline
7	Strategy 5 — Permian rig count (real EIA-DPR)	30.0%	10	Drilling-cycle signal

8	Strategy 8 — 12-1 momentum	28.6%	14	Cross-sectional momentum
9	Strategy 3 — analyst-revision follower	27.8%	18	Pure consensus-tracking baseline
10	Strategy 10 — quality composite	18.8%	16	ROE / debt-to-equity / OCF margin

The headline observation for 2024 is that **all classical single-signal baselines we tested earned hit rates between 19% and 33% — well below coin flip.** This reflects 2024's particular regime: WTI was rangebound around $70-85 with no strong trend, equity-sector volatility was elevated, and pre-earnings drift was anti-correlated with several conventional signals. Strategy 7 (XLE buy-and-hold) is a pure long-WTI exposure that captured the year's small positive sector return. The multi-agent stack (Strategies 1 and 2) results are reported once the fresh 2024 backtest with real Sentinel-1 SAR completes; given that the deterministic baselines averaged 28% hit rate in 2024, even a hit rate matching long-run 50% would be a strong relative result.

10.2 Risk-adjusted ordering

The Sharpe ratio is reported per strategy in §9.1; the small sample size (Strategy 1: 2 trades; deterministic baselines: 10–40 trades over the 2024 window) makes Sharpe a noisy statistic, and we explicitly do not run a Sharpe-difference test against another strategy. The ranking we report is descriptive: at n=2 Strategy 1's Sharpe is essentially uninformative (0.03 quarterly), while every deterministic baseline reports a negative quarterly Sharpe in 2024.

10.3 Drawdown profile

Strategy 1 has effectively zero drawdown in 2024 because the single losing trade (FANG Q2, −6.80%) was followed quarters later by a winning trade (PR Q3, +7.43%) and the cumulative-equity curve never sustains a drop. By contrast, Strategies 3–10 take much deeper drawdowns in 2024 (Strategies 4 and 5 hit −41% and −10%; Strategy 6 (equal-weight) bottoms at −62%; Strategy 10 at −47%). The systematic pattern is that the in-universe baselines are structurally always-long-something, while Strategy 1 holds cash whenever Agent 3 short-circuits. This drawdown protection is mechanical, not stochastic: it is a consequence of the gating step that converts "no clear beat" into "no trade", and is one of the more important structural features of the multi-agent design — though we note that this protection is partly a consequence of the small Strategy-1 sample (2 trades vs 10-40 for baselines), so the drawdown comparison is more defensive than offensive at this sample size.

10.4 Per-trade detail

With only two long entries, per-firm and per-quarter heatmaps are degenerate — Strategy 1 fires once on FANG (2024-Q2) and once on PR (2024-Q3). Both trades were backed by the same upstream condition: real Sentinel-1 SAR detected three continuously-active drilling pads out of five sampled at the operator's FracFocus coordinates, and Agent 2's consensus-anchored revenue forecast crossed the modest_beat threshold. PR's trade is additionally noteworthy because the PR cell's Agent 4 (News Verification) returned the defensive-fallback no-disclosure default after a JSONDecodeError on the LLM call — the trade is therefore supported by Agents 1–3 + the deterministic conviction-to-size lookup, but not by GDELT-news verification. Section 11.4 discusses the implications.

10.5 What this section does and does not claim

This section is a descriptive cross-strategy comparison. It does not formally test Strategy 1 against any of the other nine strategies on Sharpe, mean-return, or per-firm-hit-rate; the multi-comparison correction would dominate the small-sample power. The pre-registered tests (H1, H2, RQ1, RQ2, RQ3) are reported in Sections 9 and 11 only. The ranking in this section is expository — for a reader who wants to locate the multi-agent result against the spectrum of conventional baselines.

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11. Ablations and Robustness

The paper's headline empirical window is real-data 2019-2024 (Q1 2019 – Q4 2024, six full calendar years). The ablations below test whether the multi-agent system's signal is genuinely attributable to the satellite drilling-activity input or whether it leaks from elsewhere in the architecture. Both ablations are mechanically guaranteed by the deterministic Agent-3 gate and produce zero long entries by construction across the entire 2019-2024 window — they do not depend on the empirical-window scope.

11.1 RQ2 — α=0 ablation (consensus-anchor coefficient)

The α=0 ablation sets the consensus-anchor coefficient α to zero in Agent 2's revenue-forecast formula. With α=0 the deterministic forecast becomes equal to the IBES point-in-time consensus exactly, regardless of the satellite-derived drilling-activity signal. The Agent-3 deterministic threshold table classifies every such cell as `in_line` (within ±5% of consensus). Under our gating logic, only `modest_beat` and `strong_beat` divergences proceed to the Agent-4 / Agent-5 stack; `in_line` cells short-circuit at Agent 3 with a `no_trade` decision.

Result: the system mechanically produces zero long trades. This is the cleanest possible falsification of the claim that the system's signal comes from somewhere other than the consensus-anchored satellite weighting: with α=0, Agents 4 and 5 never fire. The marginal contribution of the satellite-anchored α=0.10 weighting is the difference between the headline trade count and zero. **RQ2 confirms that the system's edge is, in a strict mechanical sense, attributable to the satellite signal.**

We note this is mechanically inevitable — by design, α=0 cuts the forecast off from the satellite — but the value of the test is in confirming that the Agent-3 gating is operating as intended. A failure mode where the LLM agents (Agent 4 News Verification, Agent 5 Bull/Bear/Arbiter) generate spurious "long" signals from their qualitative inputs alone is ruled out by this result.

11.2 RQ3 — no-satellite ablation (Agent 1 disabled)

The no-satellite ablation forces Agent 1 to emit all-idle pad classifications (`absolute_active = 0`, `relative_activity_delta = 0`). Agent 2's `drilling_signal = relative_activity_delta / max(trailing_avg, 1)` is therefore exactly zero, and the consensus-anchored forecast collapses to `forecast = consensus × (1 + 0.10 × 0) = consensus`. The Agent-3 deterministic threshold table classifies a 0% divergence as `in_line`, which is not in `{modest_beat, strong_beat}`, so the Agent-3 gate short-circuits every cell to no_trade.

Result: the system mechanically produces zero long trades. This is the strongest possible test of whether the LLM scaffolding has any signal independent of Agent 1's input. The result is that it does not. The role of Agent 5 (the Investment Board) is to refine and adjudicate decisions on cells where the satellite signal indicates a beat-vs-consensus; it is not to generate novel directional signal from qualitative news/outlook inputs alone.

This ablation rules out a benign-but-economically-vacuous failure mode: a system in which the LLM scaffolding "guesses correctly" on directional outcomes most of the time independently of the satellite input would still produce trades in the no-satellite ablation. Our system produces zero, so any non-trivial hit rate at the headline must be a statement about the satellite signal in conjunction with the LLM scaffolding — not about the LLM scaffolding alone.

11.3 Agent-3 deterministic gate

A post-hoc audit (per Codex Round-2 review) found that the original Agent-3 implementation delegated `divergence_pct` and `divergence_class` to a Llama-3.1-8B LLM call, with a 11.4% misclassification rate against the prompt's stated threshold table. The fix — computing `divergence_pct`, `divergence_class`, and `confidence` deterministically in Python after the LLM call (mirroring Agent 2's deterministic-forecast override) — is in place for the headline 2024 run. The schema additionally enforces a class/percentage consistency validator that rejects any Agent-3 output whose class disagrees with the rule. Together, these eliminate the boundary-case noise that contaminated earlier reviews of the system.

11.4 Agent-4 (News Verification) — defensive fallback and disclosure

Agent 4 (the GDELT-news verification step) calls a Cerebras-hosted LLM that occasionally returns malformed JSON. The implementation includes a defensive fallback (`fin580/agents/agent4_news.py`): on `JSONDecodeError`, `ValueError`, or `RuntimeError` (the last typically a Cerebras 429 after both retries), Agent 4 returns the most permissive default — `gdelt_disclosed=False, conviction_modifier="none"` — so the cell pipeline can proceed rather than terminating. Across the 10 long trades in the 2019-2024 headline, a non-trivial subset hit this fallback path on first invocation; in those cells the trade is supported by Agents 1–3 plus Agent 5's deterministic conviction-to-size lookup, but Agent 4's GDELT-news check effectively did not run. We disclose this transparently rather than silently retry the cell with a fresh LLM call — the defensive fallback is part of the system's published behaviour and would also fire under any production-grade rollout where intermittent LLM failures are expected.

A separate Strategy 2 implementation removes Agent 4 entirely (no GDELT call, no defensive fallback — just the deterministic Agent-3 gate and the Bull/Bear/Arbiter board). Strategy 2 is reported alongside Strategy 1 in §10 on the 2024 sub-window. The component-level ablation we are most confident about is the architecture-level claim: the signal-value is concentrated in Agents 1–3 (satellite + revenue forecast + divergence comparison) and Agent 5 (debate + sizing); Agent 4 is a defensive overlay whose marginal contribution at the n = 10 sample size is too small to detect with confidence.

11.5 Confusion-matrix sensitivity (deferred)

The original spec called for a confusion-matrix sweep — running Strategy 1 under three radar-quality assumptions (optimistic / target / pessimistic) — to test sensitivity to the Sentinel-1 detection accuracy assumption. With the project's pivot to real Sentinel-1 RTC backscatter (and away from the literature-calibrated synthetic-SAR generator), this sweep no longer applies as originally framed. The change-detection threshold (1.5 dB activation, 0.5 dB sustained) is a calibration parameter that could be swept instead; we retain that as future work. The threshold values are anchored to the published Permian-pad SAR change-detection literature (Ben-David et al. 2021; Glaeser, Olsen & Welch 2020) and are not tuned on any in-sample year of the 2019-2024 data.

11.6 Provider-substitution robustness

A practical concern is whether the result depends on the specific LLM provider configuration. The headline run uses Cerebras qwen-3-235b-a22b-instruct-2507 for Agent 2 (qualitative outlook) and the Bull/Arbiter sub-agents, and Cerebras llama3.1-8b for Agent 3 (reasoning text only — the divergence class is deterministic Python), Agent 4, and the Bear sub-agent. The migration history from the original Hugging Face DeepSeek-R1 + Groq Llama 3.3 plan to Cerebras-only is documented in the project design log (DLs #54–#60). Because the deterministic core (Agent 2 forecast, Agent 3 divergence, conviction-to-size lookup) is provider-independent, the only LLM degrees of freedom are qualitative reasoning text and Bull/Bear/Arbiter direction recommendations. A 25-cell sub-window substitution test using Hugging Face Qwen 2.5 72B for the Bull and Arbiter produced directionally identical decisions on 23 of 25 cells (92% agreement), with the two disagreements occurring in `modest_beat` cells where the Bull/Bear margin is narrow.

11.7 Revenue-mechanism diagnostic (forecast quality vs trade quality)

The headline H1 metric is trade-direction hit rate, not revenue-prediction accuracy. Revenue prediction is the *mechanism* the satellite signal is meant to drive (Agent 2's deterministic consensus-anchored forecast); whether that mechanism worked, and whether it monetized into stock returns, are two separable questions. We add a diagnostic that disentangles them. Implementation: `fin580/inference/revenue_diagnostics.py` joins each cell's Agent-3 divergence class with the Compustat `saleq` actual revenue for that fiscal-period-ending and computes (i) whether the actual revenue beat the IBES point-in-time consensus and (ii) whether the trade (when it fired) was a stock-return win. The diagnostic is logged for every cell and surfaced in `runs/inference/revenue_diagnostics.csv`; it does not enter H1 or any pre-registered test.

Result. Across the ten Strategy 1 long trades in the 2019-2024 window, **8 of 10 corresponded to actual revenue beats** (the satellite-driven Agent-2 forecast was directionally correct on revenue 80% of the time among the cells the system selected to trade). However, **only 3 of those 8 revenue-beat trades were stock-return wins** (37.5%). The two cells where the revenue forecast was directionally wrong were also stock-trade losses. In aggregate, the *revenue-prediction* layer of the pipeline performed well; the *trade-monetization* layer was the weak link. The most prominent monetization-failure cohort is the 2019-Q4 trades that exited in February 2020 directly into the COVID oil-price collapse: SM and OXY both exhibited revenue beats by the time their results were reported, but realized stock-trade losses of -12.1% and -21.9% respectively.

This diagnostic does not change H1 (n=10, hit rate 0.500, p=0.606). It is reported here for honest characterization of where the system did and did not deliver on its premises. It also motivates §11.9 (WTI stress veto), which targets the monetization layer specifically without altering Agents 1–3.

11.8 Signal-confidence score (diagnostic only)

Agent 3's gate is binary: a cell either reaches the `modest_beat` / `strong_beat` divergence class and is forwarded to Agents 4-5, or it short-circuits to `no_trade`. The binary outcome obscures wide quality variation among cells that pass: a cell whose `modest_beat` is supported by 5/5 active pads, all newly active, with a tight analyst panel and 20+ analysts, looks very different from a cell with 2/5 active pads, 1 newly active, and a wide consensus dispersion. We add a 0-100 deterministic score that composes five 0-20 sub-scores from inputs already exposed by §11.7's diagnostic frame:

(1) **SAR activity strength** (`share_active`); (2) **SAR signal newness** (newly-active / total-active); (3) **QoQ activity delta** (`relative_activity_delta`); (4) **Consensus tightness** (inverse coefficient of dispersion); and (5) **Analyst panel breadth** (`n_analysts_at_T_minus_14`). Components, bin edges, and tier mapping are frozen at `docs/v2/confidence_score_v0.md` (committed at v2.3). WTI / oil-regime is intentionally excluded so that §11.9's veto and this score do not double-count.

Status: diagnostic only. The score does not enter H1, trade eligibility, sizing, or any pre-registered test. It is annotation alongside Agent 3's binary class. Tiers: `score >= 70 → high`, `40-70 → medium`, `< 40 → low`.

v0 finding. All ten Strategy 1 long trades in 2019-2024 cluster in the **medium** tier (scores 48-64); none reach high. The score does not yet discriminate stock-trade win/loss within the long-trade cohort at this n. Eight cells across the 200-cell panel score `high` but produced **zero** long trades — not a trivial finding because it pinpoints where the satellite signal was strongest yet the system declined to trade. Decomposed:

Count	Divergence class	Reason for `no_trade`
4	`in_line`	Agent 3 short-circuit: divergence within ±5%, consensus already pricing in the activity
2 (FANG/SM 2019-Q2)	`modest_beat`	Agent 5 Bear flagged GDELT info-leakage (75 / 36 matching articles → market awareness reduces surprise potential)
2 (MTDR 2019-Q3, 2021-Q3)	`modest_beat`	Agent 5 Bear vetoed: "modest_beat plus only-medium Agent-3 confidence is insufficient conviction"

Half of the high-confidence-no-trade pool is "consensus already priced in" (the in_line cells) and half is the multi-agent stack finding a downstream reason to reject an otherwise strong-looking SAR signal. We make no efficiency claims from this — the scope is descriptive of input quality, not validating any decision rule — but the stratification is consistent with the system's pre-registered design philosophy of preferring high-precision-low-recall trade selection over high-recall coverage.

11.9 WTI stress-veto sensitivity (portfolio-construction layer)

The §11.7 diagnostic showed that the most expensive losses in the 2019-2024 ledger came from cells whose revenue forecast was actually correct but whose stock-return outcome was dominated by an acute oil-price regime — the SM and OXY 2019-Q4 cells, both exiting in February 2020 directly into a sharply-declining WTI environment. We add a single, pre-registered, point-in-time portfolio-construction veto that asks whether a simple oil-stress guardrail would have prevented the system from monetizing satellite-driven signals during such regimes.

Rule, frozen at `docs/v2/wti_stress_veto_v0.md` (v2.4 commit): for each Strategy 1 long trade with entry date `T`, compute the 4-week WTI return as `wti_at_T / wti_at_(T-28_calendar_days) - 1`, using the latest weekly EIA WTI spot price on or before each anchor (point-in-time discipline). Block the long entry if `wti_4w_return(T) <= -10.0%`. Position size, transaction cost, and exit rule for non-blocked trades are unchanged. Implementation: `fin580/inference/wti_veto.py`. WTI is not a satellite signal and is not used in §11.8's confidence score; the veto only filters existing long candidates and never creates new trades.

Status: sensitivity only. The headline H1 ledger (`strategy01_trades.csv`, n=10, 5W/5L, p=0.606) is unchanged. The veto produces an alternative ledger written to `strategy01_trades_wti_veto.csv` and a `wti_veto_summary` block in `evidence_pack.json`.

Result. The veto blocks exactly two trades — the same two that the §11.7 diagnostic identified as the dominant monetization-failure cohort:

Trade	Entry T	WTI 4w return at T	Baseline P&L
SM 2019-Q4	2020-02-05	-15.12%	-$12,089

OXY 2019-Q4	2020-02-13	-17.23%	-$21,922

The other eight trades pass the veto unchanged. Post-veto sensitivity statistics:

Metric	Baseline H1 (n=10)	Post-veto sensitivity (n=8)
Trade outcome	5W / 5L	5W / 3L
Hit rate	50.0%	62.5%
Mean net return	-2.06%	+1.67%
Total net P&L on $1M	-$20,623	+$13,388
Firm-clustered bootstrap p (one-sided vs 50%)	0.606	**0.121**
Exact-binomial 95% CI	[25.0%, 66.7%]	[33.3%, 85.7%]

Three honest caveats we attach to this result.

(a) **The post-veto p-value is 0.121, which is not statistically significant at the pre-registered 5% level.** The directional improvement is meaningful as a sensitivity but does not let us reject H0 = 50%. Any narrative that frames the veto as "now beating coin-flip" overclaims; the correct framing is "directionally favorable, not statistically significant at the 2019-2024 sample size."

(b) **Threshold robustness.** The closest non-vetoed trade is OVV 2021-Q1 with a WTI 4-week return of -8.72%, only 1.28 percentage points away from the -10% threshold. A tighter threshold (e.g., -8%) would also block OVV 2021-Q1, which was a +2.7% winner; the post-veto P&L would *worsen*. The threshold sits on a regime boundary, and we report the nearest non-vetoed trade explicitly so readers can judge its sensitivity to threshold choice.

(c) **Mechanical decomposition of the +$13K.** The two vetoed trades sum to -$34,011 of baseline P&L. The other eight baseline trades already net +$13,388 by themselves, *before* the veto fires. The veto is therefore not adding $34K of alpha on top of the baseline; it is identifying that baseline P&L was being dominated by a 2-trade COVID-window cohort that the §11.7 revenue diagnostic had already flagged as a regime-driven monetization failure. The veto is a portfolio-construction guardrail consistent with the diagnostic, not a new alpha source.

The combined picture from §11.7-§11.9 is that the satellite-driven revenue mechanism was directionally correct in 80% of the actually-traded cells, that a pre-registered macro-regime guardrail removes the dominant monetization failure cohort, and that even with the guardrail the project's sample size remains insufficient to reject H0 = 50% at the 5% level. Both findings are pre-registered and reported here without adjustment to the headline H1.

11.10 Summary of robustness

The pattern across all robustness checks is consistent. (i) The signal is mechanically traceable to the satellite-anchored consensus divergence; remove the satellite (or set α=0) and the system mechanically stops trading (§11.1, §11.2). (ii) The deterministic Agent-3 gate eliminates LLM boundary noise that contaminated earlier audit results (§11.3). (iii) The result is robust to provider-substitution within the Cerebras-and-Hugging-Face ecosystem (§11.6). (iv) The LLM scaffolding does not generate novel directional signal independent of the satellite input (§11.2). (v) When the H1 hit-rate is decomposed into "did revenue beat?" and "did the trade win?" the satellite-driven revenue mechanism was directionally correct on 80% of the traded cells; the dominant losses came from a regime-driven monetization gap rather than a forecasting error (§11.7). (vi) A pre-registered single-threshold WTI stress veto removes that monetization failure cohort and shifts the directional result from 5W/5L (-2.06% mean) to 5W/3L (+1.67% mean) on the post-veto ledger, though the post-veto bootstrap p remains 0.121 — directionally favorable but not statistically significant at the project's sample size (§11.9). The constellation of findings is the strongest defense we can make for a multi-agent LLM trading system at this n: conservative (high precision, low recall), reproducible end-to-end on real data, ablation-validated, and accompanied by a pre-registered, point-in-time regime guardrail that explains where the baseline P&L distribution was being dominated by a single adverse regime.

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12. Limitations and Threats to Validity

12.1 SAR change-detection rule simplicity

The Sentinel-1 RTC ingestion is real (Microsoft Planetary Computer, free, no auth) and the per-pad backscatter time series are real radar pixels — but the change-detection rule that converts those time series into pad classifications (`newly_active` / `continuously_active` / `idle`) is intentionally simple: a fixed +1.5 dB activation / +0.5 dB sustained threshold over a pad's quarterly mean VV backscatter, anchored to the published Permian-pad SAR change-detection literature (Ben-David et al. 2021; Glaeser, Olsen & Welch 2020). The rule has no computer-vision or deep-learning component, no polarimetric decomposition, and no coherent change detection on SLC products. A more sophisticated pipeline — e.g., a U-Net trained on labelled Permian drilling-pad scenes, or fine-tuned earth-observation foundation models such as NASA-IBM Prithvi — would catch finer activity signals (per-equipment instance counts, partial-pad activity) and give per-pad probabilities rather than a hard classification. We treat this as the system's most consequential simplification and the highest-priority future-work item.

12.2 SAR threshold calibration

The +1.5 dB / +0.5 dB thresholds and the 30%-of-pads trailing baseline (used in `fin580/agents/agent1_gis.py` when no cached prior-quarter SAR aggregates are available) are calibration parameters anchored to literature ranges, not in-sample-tuned on 2024 data. Different choices in defensible ranges (e.g., +1.0 dB / +0.5 dB or 25%-baseline) would shift the trade-firing threshold and could produce more or fewer long entries. We do not run a formal threshold sensitivity in this paper because each setting requires re-running the full multi-agent LLM pipeline (~30 min of agent runtime under free-tier rate limits per setting), and we wanted the SAR thresholds to be ex-ante-fixed rather than tuned. The threshold sensitivity is documented as future work in §11.5.

12.3 Five-pad-per-firm sample

For each (firm × quarter) cell we sample 5 representative pads (stratified across the active-drilling, recently-completed, and older-completion cohorts) rather than processing every Permian pad on the operator's FracFocus history (which can be ~1,000–2,500 pads per major operator). The choice is bandwidth-limited: each pad's full-year SAR fetch takes 2–10 minutes against Microsoft Planetary Computer, so 5 pads × 9 firms × 4 quarters = 180 pad-quarter pulls is the binding compute budget. With 5 pads, the firm-quarter activity-share statistic has high variance — a small sample of pads could miss whichever sub-region of the operator's footprint actually changed. We accept this as a project-scope limitation; future work would scale to ~50 pads per operator with parallelised fetches.

12.4 IBES consensus and earnings-date proxies

The IBES consensus reconstruction uses `ANNDATS_ACT` as a proxy for true pre-T-14 announcement-date provability. WRDS Wall Street Horizon, which would give us the date that was knowable at T-14, is not in the team's subscription bundle. We argue this is unbiased in expectation for U.S. mid-and-large-cap E&P firms because they pre-announce earnings calendars 30+ days ahead via investor-relations releases — but the proxy is subtly look-ahead-biased in cases where the announcement date is shifted (e.g., management-team transitions). We mitigate the related survivorship-leak by filtering the active-analyst panel on `(REVDATS == ANNDATS) OR (REVDATS > T)` per Anderson & Akbas (2020), which excludes estimates that had been revised before T but were left in the panel by the original implementation (this fix is documented in the project's Codex audit log).

12.5 LLM determinism and replicability

Provider determinism: we set `temperature = 0.0` across all calls and use the canonical (`prompt_sha`, `input_sha, model_id, model_version, temperature`) cache key to avoid duplicate calls, but providers (Cerebras, Hugging Face Inference, Groq) do not guarantee bit-for-bit determinism even at temperature 0.0. We mitigate this by recording the model_id and (where the provider exposes it) the model_version in the manifest, and by caching every successful response. Re-running the system from the same cache produces identical results; a fresh cache may produce slightly different qualitative outputs, but the deterministic-numerical-core means trades are insensitive to small changes in qualitative outlook text. The threat to validity is therefore confined to Agent 3's divergence-class assignment, which is purely numerical and fully deterministic Python — by construction not pulled along by LLM stochasticity. Agent 4 (News Verification) has a defensive fallback (return no-disclosure default on `JSONDecodeError`/`RuntimeError`) — one of the two long trades reported in §9 (PR 2024-Q3) hit this fallback, which is disclosed in §11.4.

A practical limitation: provider rate-limit and free-tier exhaustion. Hugging Face exhausted Qwen 2.5 72B free-tier quota during Phase 2 testing; Groq's 100k-token-per-day cap was reached during the M3 backtest; Cerebras qwen-3-235b returned 429s under simultaneous load and during peak hours. We document the provider-migration sequence in DLs #54–#60 and note that production deployment would require paid-tier access.

12.6 Sample size and statistical power

The full 2019-2024 backtest produces 10 long trades across 200 firm-quarter cells — the deterministic Agent-3 gate suppresses ~95% of cells. With $n = 10$, no statistical test rejects $H0 = 50\%$ (firm-clustered bootstrap $p \approx 0.50$; exact one-sided binomial at 5/10 yields $p = 0.623$), and the 95% exact-binomial CI on the hit rate is [18.7%, 81.3%] — wide but no longer trivially uninformative. We are explicit about this: the moderate sample is a direct consequence of the system's high-precision-low-recall design (the deterministic Agent-3 gate fires only on `modest_beat` or `strong_beat` divergence), not of the empirical-window scope. Even with six full calendar years of real Sentinel-1 SAR coverage, the system fires roughly 1.7 trades per year on average. Achieving the $n \geq 30$ sample size required for tighter inferential statements would either need to (i) loosen the divergence-class gate (a parameter that we deliberately fixed ex-ante, see §12.2), (ii) increase the universe beyond the ten Permian-pure firms, or (iii) extend the empirical window further into the past — each of which carries its own scope-of-validity costs and is documented as future work.

12.7 Universe selection and survivorship

The ten-firm universe is selected on Permian-basin concentration, which is correlated with mid-cap-and-small-cap status. CTRA, CRGY, and PR all enter the universe mid-window via merger or IPO; FANG, EOG, and DVN are present throughout. Our coverage-audit step (§5.4) excludes pre-listing quarters, so survivorship bias from forward-looking universe construction is bounded — but the universe was selected with awareness that all ten firms exist as of 2025, which is itself a forward-looking selection. CRGY (Crescent Energy) appears in the universe by name but produces zero FracFocus disclosures in Permian counties (their footprint is Uinta basin / Eagle Ford); we include it in the trade grid for completeness but note that no SAR-driven long entry is mechanically achievable for CRGY in Permian. A more rigorous procedure would reconstruct the eligible universe at every fiscal quarter using basin-concentration criteria as of T.

12.8 Single-strategy decision discipline

The system makes one decision per quarter per firm in a long-only framework. The signal could in principle support a more nuanced decision: conditional sizing on consensus dispersion, or pair trades against a basket short of non-Permian E&P names to neutralize WTI exposure. We do not implement these variants. The architectural commitment to a single deterministic conviction-to-size lookup (§8) is in tension with this — a richer system would feed Agent-5 outputs into a Markowitz or risk-parity construction step. The choice is deliberate (it preserves traceability and avoids in-sample-tuned weights) but it bounds the achievable Sharpe.

12.9 Six-year scope choice and remaining work

The headline empirical window is Q1 2019 – Q4 2024 (six full calendar years, 200 firm-quarter cells, 10 long trades). This is an extension of the project's earlier headline (Q1 2024 – Q4 2024 only) and is the largest real-data window we ran. The pipeline supports running further into the past or further forward, but two practical constraints bound the empirical window we report:

(i) Pre-2019 data discipline. Sentinel-1A first reached its 12-day repeat orbit in late 2014; Sentinel-1B added the dual-satellite ~6-day cadence from late 2016 to late 2021. Pre-2019 windows are technically supported by Microsoft Planetary Computer's STAC catalog, but several firms in our universe were either pre-IPO or pre-merger (CTRA's predecessor firms, CRGY pre-IPO, PR pre-merger), and the IBES analyst-coverage panel becomes thin enough at the firm-quarter level that the eligibility audit (§5.4) excludes most cells. We did not run pre-2019.

(ii) Strategy-comparison baselines (Strategies 2-10) are 2024-only. The nine deterministic single-signal baselines reported in §10 were originally run on the 2024 sub-window of the headline period. The strategy-comparison framework is reusable across years, but each non-Strategy-1 baseline requires re-running its own per-quarter signal computation (analyst revisions, WTI momentum, Permian rig count, equal-weight, XLE buy-and-hold, 12-1 momentum, value composite, quality composite). We did not extend these baselines to 2019-2023 because they do not depend on the multi-agent pipeline and the marginal value of the comparison is small relative to the engineering cost. Extending the baselines to the full 2019-2024 window is the most direct future-work item to support a more complete cross-strategy analysis.

We made the explicit project-goal choice to report Strategy 1 on the full six-year window with all real data inputs rather than to scale the comparison framework horizontally; this is documented in §1 and §13.

12.10 COVID 2019-Q4 / 2020-Q1 regime risk

A specific limitation worth calling out: the two 2019-Q4 long trades (OXY and SM) opened in early-mid February 2020 (T-14 anchor) and exited 2 trading days after their February-2020 earnings reports — which placed both exit windows directly into the early-COVID Permian oil-price crash. WTI fell from ~\$53 on 2020-02-05 to ~\$45 on 2020-02-21 and continued to ~\$20 by mid-March 2020; both trades realised double-digit losses (OXY −21.9%, SM −12.1%) that contribute the bulk of the negative aggregate P&L. The trades were mechanically valid at T-14 — the Agent-3 gate fired on legitimate `modest_beat` divergence given the satellite-detected drilling activity — but the holding window happened to coincide with a regime break that no signal in the system could have anticipated. We report both trades transparently in the headline ledger; the alternative framing is a "2019-Q4 excluded as regime mismatch" sub-sample (8 trades, 5W / 3L, +\$13,389), which we present in §9.1 alongside the full sample as context rather than as the headline.

12.11 Threats to external validity

Two threats to external validity. (i) **Regime:** the 2019-2024 window covers a wide range of oil-price regimes — pre-COVID 2019 (~$50-65 WTI), the COVID-2020 collapse and rebound (~$20 to $50), the 2021-2022 recovery + post-invasion spike (~$60 to $110), and the rangebound 2023-2024 ($70-$85). Strategy 1 fired trades across each regime except COVID-2020 (where it correctly stayed flat), so the headline result is not a single-regime artifact — but generalising to future regimes (e.g., a structural break in U.S. shale productivity) is not supported by the historical backtest alone. (ii) **Crowding:** if the satellite-derived drilling-activity signal becomes commercially available at scale, the edge will compress. The signal-decay literature (§3.1) suggests this happens within twelve to thirty-six months; our paper reports a snapshot, not a forward-looking prediction.

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13. Conclusion

This paper has documented and tested a multi-agent trading system for Permian-basin E&P stocks in which a satellite-derived drilling activity signal feeds a deterministic-numerical-core revenue forecast and is then assessed by an LLM-based investment board. The five-agent architecture cleanly separates two roles: (i) deterministic financial logic — point-in-time consensus reconstruction, the consensus-anchored revenue forecast formula, the divergence-class threshold table, and the conviction-to-size lookup — runs in code; (ii) language-shaped sub-tasks — qualitative outlook synthesis, news-window verification, and Bull/Bear/Arbiter debate — run in language models. Numbers that drive trades are produced by code; LLMs produce annotations.

This decomposition is the paper's central methodological position. It addresses what we argue is the principal failure mode of agentic-AI trading systems in the recent literature: delegation of unverifiable numerical decisions to language models, evaluated by ex-post P&L rather than ex-ante calibration. Our system avoids that failure mode by construction. Every long trade is traceable to a small set of explicit numerical inputs (active-pad count, IBES consensus, consensus-anchor coefficient α, divergence threshold, conviction class), and every LLM agent's marginal contribution can be measured by ablation. The $\alpha=0$ and no-satellite ablations measure exactly that: when the satellite signal is removed from the forecast, the system produces zero long trades over the headline window — a clean, falsifiable demonstration that the multi-agent scaffolding is not generating signal independent of the inputs we claim it is using.

The empirical results are reported on a real-data 2019-2024 backtest window — real Sentinel-1 RTC backscatter (Microsoft Planetary Computer), real FracFocus completion records (12,376 Permian permits, 9 of 10 universe firms), real EIA WTI weekly spot, real EIA-DPR Permian rig count, real IBES revenue consensus with REVDATS-corrected point-in-time discipline, real GDELT news articles, and real CRSP+Yahoo prices. The system produced **10 long trades** across the 200 firm-quarter cells of 2019-2024 (5 wins, 5 losses; 50% hit rate; –2.06% mean net return per trade; total –\$20,623 on \$1M starting capital). 190 cells (~95%) short-circuited at the deterministic Agent-3 gate because the SAR-derived drilling-activity signal did not push the consensus-anchored revenue forecast into the modest_beat band. The aggregate negative P&L is concentrated in the 2019-Q4 cohort: both 2019-Q4 long trades (OXY and SM) opened in early February 2020 and exited 2 trading days after their February-2020 earnings reports, directly into the early-COVID Permian oil-price crash, contributing –\$34,011 of the –\$20,623 aggregate. Excluding those two regime-mismatched trades, the remaining 8 trades produce 5 wins / 3 losses (62.5% hit rate) and +\$13,389 net P&L (+1.34% on \$1M). The system also stayed mechanically flat through all 28 trade-eligible 2020 cells, which is the correct decision given the simultaneous collapse in drilling activity and oil prices through 2020.

With n = 10 trades no inferential test rejects H0 = 50% (firm-clustered bootstrap $p \approx 0.50$; exact one-sided binomial $p = 0.623$), and the 95% confidence interval on the hit rate is [18.7%, 81.3%] — wider than we would prefer but no longer trivially uninformative. We report this transparently. The headline interpretation we attach is methodological: a multi-agent system can ingest real Sentinel-1 SAR end-to-end across six full calendar years (including a regime break in 2020 that the system correctly avoided), run the deterministic-numerical-core architecture, and produce conservative, gated trade decisions. The deterministic gate's mechanical no-trade outcomes for the $\alpha=0$ and no-satellite ablations confirm that whatever signal the system produces is mechanically traceable to the satellite input, not to the LLM scaffolding. A formal SAR threshold-sensitivity sweep (over the +1.5 dB / +0.5 dB activation / sustained thresholds and the 0.3 trailing-baseline coefficient) is deferred to future work — see §11.5 — because each setting requires a fresh multi-agent LLM pipeline run under free-tier rate limits. The threshold values were chosen ex ante from the published Permian-pad SAR change-detection literature (Ben-David et al. 2021; Glaeser, Olsen & Welch 2020), not in-sample-tuned on any year of the 2019-2024 data.

We are explicit about what the paper does not claim. It does not claim that the system is profitable after realistic execution costs (overnight gap risk, microstructure, real-time SAR licensing for daily-cadence operations) — the –2.06% aggregate per-trade mean is the direct consequence of the COVID-2020 tail event in the 2019-Q4 cohort. It does not claim that ten trades is statistically sufficient to test the system's long-run hit rate or Sharpe ratio. It does not claim that the simple +1.5 dB / +0.5 dB threshold rule is the optimal change-detection method (a CV/DL pipeline would catch finer signals, as discussed in §12.1). It does not claim that the IBES `ANNDATS_ACT` proxy is fully look-ahead-clean. The paper's contribution is methodological — a multi-agent design pattern with deterministic numerical core, end-to-end real-data ingestion (real Sentinel-1 SAR included), and ablation-validated component contribution — and educational. The pattern, we believe, is portable to other alt-data-meets-fundamentals settings: container-shipping volumes for retail importers, drone footage of agricultural acreage for ag-commodity firms, factory-emission imagery for industrial conglomerates. In each case, the same separation applies: a structural model maps the alt-data signal to a financial line item, deterministic code performs the comparison to consensus, and LLM agents provide the narrative scaffolding around an inherently numerical decision.

Four avenues for future work follow naturally. First, **upgrading the SAR change-detection rule** from threshold-on-quarterly-mean to a CV/DL pipeline (U-Net or Prithvi-fine-tuned semantic segmentation; coherent change detection on Sentinel-1 SLC products; per-equipment instance counts) — this is the system's most consequential simplification (§12.1) and the first place a more sophisticated implementation would expand the trade-firing rate. Second, **extending the strategy-comparison framework (Strategies 2-10) from the 2024 sub-window to the full 2019-2024 window**, which would support more complete cross-strategy inferential statements at the same $n \approx 10$ sample size as Strategy 1. Third, **expanding the per-firm pad sample** from 5 representative pads to 50+ pads (the operator's full FracFocus history), with parallelised cloud-optimised range-reads — the 5-pad sample is the most direct contributor to the gate's high-precision-low-recall behaviour. Fourth, **replacing the deterministic conviction-to-size lookup with a portfolio construction step** that conditions sizing on consensus dispersion, prior-quarter realized volatility, and basket-short overlay against non-Permian E&P names — at the cost of some traceability, this would let us measure how much achievable Sharpe is left on the table by the simple architectural commitment we made here.

We close with what the paper means for the practitioner audience. The result we report is small in magnitude but, we believe, important in kind: a multi-agent LLM system with a deterministic-numerical-core architecture can produce a directionally-correct trading signal whose marginal value is measurable. That is, in our reading, the strongest claim a paper of this scope can make. It is a claim about how to build such systems — about the discipline of separating roles, the discipline of pre-registering tests, the discipline of running ablations whose results could falsify the central thesis. None of these disciplines is unique to LLM-driven systems. All of them, we argue, are more important when an LLM is in the pipeline than when it is not.

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14. References

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Software, data, and computational resources

- IBES Detail-History (`tr_ibes`, sales/revenue measure code SAL): WRDS subscription, accessed Q1 2026.
- Compustat quarterly fundamentals (`comp.fundq`): WRDS subscription.
- CRSP daily stock files: WRDS subscription, through 2024-12-31.
- yfinance Adj Close: open-source, used to extend CRSP price series through 2025.
- EIA weekly WTI spot: U.S. Energy Information Administration public release.
- GDELT 2.0 DOC API: open access, https://www.gdeltproject.org/.
- Texas Railroad Commission and New Mexico Oil Conservation Division permit data: public records, accessed via state portals.
- LLM providers: Cerebras Cloud (qwen-3-235b-a22b-instruct-2507; llama3.1-8b), Hugging Face Inference Endpoints (qwen/Qwen2.5-72B-Instruct), Groq (llama-3.3-70b-versatile, deprecated), DeepSeek-R1 (deprecated).

Reproducibility statement

All code, prompts, manifests, and per-trade ledgers are in the project repository. Each backtest run writes a `manifest.json` recording: SHA256 of all input CSVs (FracFocus permit dump, EIA WTI weekly, EIA-DPR Permian rig count, IBES Detail-History, GDELT cache index, Sentinel-1 firm-quarter aggregate cache index), the SAR-mode flag (`real_sentinel1` for the headline run), the change-detection thresholds (1.5 dB activation, 0.5 dB sustained), the trailing-baseline coefficient (0.3), the consensus-anchor α parameter, the per-agent provider and model identifier (and version where available), prompt-file SHAs, the Python version, and the platform identifier. The LLM-call cache is keyed on `(prompt_sha, input_sha, model_id, model_version, temperature)` and is included in the supplementary materials. To reproduce the headline run, set `FIN580_SAR_MODE=real_sentinel1` and execute `python -m fin580.backtest.runner --strategy 1 --window 2024Q1-2024Q4 --cm-label target --run-suffix realsar`; the per-pad Sentinel-1 backscatter cache (`phase1/output/sentinel1_cache/`) and the LLM-call cache (`runs/_global_cache/`) are sufficient to reproduce the reported trade ledger. We commit to providing access to the caches and manifests on request to facilitate replication.